ISLAMIC UNIVERSITY OF BAHAWALPUR · DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING
Industrial IoT & Environmental Automation Laboratory

OFFICIAL ENGINEERING SPECIFICATION & FIELD OPERATING PROTOCOL

REVERSE VENDING MACHINE (RVM) STANDARD OPERATING PROCEDURES (SOP) & FIELD ENGINEERING MANUAL

Technical Standard Operating Procedures (SOP-01 through SOP-10), Microcontroller Pinout Specifications, Cloud Sync & Offline Queueing Protocols, Diagnostic Fault Trees, and Preventive Maintenance Schedules.

DOCUMENT CONTROL & SPECIFICATION

Document ID:	**DOC-RVM-SOP-2026-V3.2**
Firmware Build:	**v3.2.0-STABLE**
Desktop Kiosk Engine:	**WPF .NET 8.0 / C# 12**
Target Controller:	**Windows 10/11 IoT Enterprise**
Microcontroller:	**Arduino Uno R3 (ATmega328P)**

DEPLOYMENT TOPOLOGY

Primary Screen:	**1080x1920 Portrait Touch**
Secondary Screen:	**1920x1080 Landscape Signage**
Local Database:	**SQL Server LocalDB (MDF)**
Central Cloud API:	**HTTPS REST / JSON FIFO Queue**
Telemetry Stream:	**Serial COM 9600 Baud / 60s Ping**

DOCUMENT REVISION HISTORY

Version	Date	Author / Engineering Group	Description of Significant Engineering Modifications
	Jan 2026	Embedded Systems Team	Initial prototype build, single screen interface, basic optical pulse sizing.
	May 2026	IoT Software Engineering	Dual-screen topology, SQL Server LocalDB offline ledger, ultrasonic drop validation.
	Aug 2026	Systems Integration Dept	Central Cloud API sync, QR code mobile wallet onboarding, telemetry terminal.
	Sep 2026	Lead Software & Industrial Team	1-min Idle Mode 50% expansion, contained zoom, zero-overlap geometry, 001 demo hotkey.

1. System Hardware & Display Topology

Architectural overview of the dual-screen industrial kiosk, computer controller, power subsystems, and failover watchdogs.

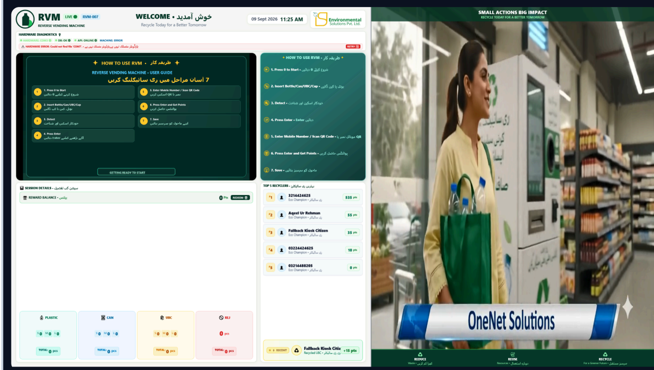
PRIMARY INTERACTIVE TOUCH KIOSK (SCREEN 1)

Resolution: 1080 x 1920 Portrait Touchscreen (55" Industrial IPS).
Host Window: MainWindow.xaml (WPF Direct3D Accelerated).
Primary Function: Drives citizen deposit guidance, live container breakdown, mobile wallet entry modal, star rating UI, and 50% expanded idle video standby.

SECONDARY PUBLIC AWARENESS DISPLAY (SCREEN 0)

Resolution: 1920 x 1080 Landscape Signage (43" Ultra-Bright LED).
Host Window: LandscapeWindow.xaml.
Primary Function: Ambient pedestrian attraction, real-time campus recycling leaderboard, daily CO2 savings aggregate, and commercial video signage reel.

SECONDARY DISPLAY PREVIEW (1920 x 1080 LANDSCAPE)



INDUSTRIAL PC & ELECTRICAL POWER SPECIFICATIONS

Subsystem	Component Specification	Operational Parameters & Engineering Margin
Industrial PC (IPC)	Intel Core i5-1135G7, 16GB DDR4 RAM, 256GB NVMe SSD	Fanless industrial chassis; operational temp range -10°C to +55°C. Windows 10 IoT Enterprise.
Power Conditioning	1000VA / 900W Online Double-Conversion Sine Wave UPS	Zero-transfer-time AC battery backup. Provides 45 minutes of complete runtime during grid brownouts.
DC Power Rails	Mean Well 12V 10A (Sensors) + 5V 5A Buck (Logic & Servos)	Independent isolated DC power supplies prevent inductive motor spikes from resetting Arduino logic.
Serial COM Interface	Industrial FTDI USB-to-UART Opto-Isolated Cable	Baud rate 9600, 8-N-1. Hardware keep-alive ping every 5000ms. Auto-reconnect thread upon disconnect.
Safety Watchdog	WPF Application Watchdog + Hardware Reset Relay	Monitors thread responsiveness; auto-restarts kiosk process if UI message pump stalls for > 15 seconds.

DUAL-SCREEN COORDINATION ENGINE

App.xaml.cs executes display enumeration via System.Windows.Forms.Screen.AllScreens. The secondary display is bound to Screen.AllScreens[0] while the primary portrait touch interface launches full-screen on Screen.AllScreens[1]. When only one display is detected, secondary display gracefully defaults to background virtual rendering.

2. Microcontroller Pinout & Sensor Wiring

Complete physical pinout assignments, signaling logic, and optical thresholds for the Arduino Uno R3 controller.

Pin	Direction	Subsystem / Sensor	Electrical Signal	Functional Logic & Trigger Conditions
D2	**INPUT**	Bottom IR Photoelectric Sensor	Active LOW (NPN NO)	Detects container entrance at intake aperture. Starts the 500ms classification timer.
D3	**OUTPUT**	Ultrasonic Transducer (Trig)	5V TTL 10µs Pulse	Emits 40 kHz acoustic burst to measure container drop velocity and baseline distance.
D4	**INPUT**	Ultrasonic Transducer (Echo)	Pulse Width (µs)	Calculates flight distance. Used for drop verification and container length profiling.
D5	**INPUT**	Inductive Proximity Sensor	Active HIGH (PNP NO)	Detects ferrous and non-ferrous metal. HIGH = Aluminium Can (UBC), LOW = PET Plastic.
D7	**INPUT**	Mid-Level IR Photoelectric	Active LOW (NPN NO)	Optical height threshold for Medium containers (> 18cm). Beam broken = Medium size.
D8	**INPUT**	Top-Level IR Photoelectric	Active LOW (NPN NO)	Optical height threshold for Large bottles (> 26cm). Beam broken = Large size.
D9	**OUTPUT**	Drop Gate Servo Motor	50Hz PWM (1000–2000µs)	0° = Closed Security Gate; 90° = Accept/Drop into bin; 180° = Reject Return to User.
D10	**INPUT**	Plastic Bin Optical Level Beam	Active LOW (NPN NC)	Infrared through-beam across 120L plastic bin rim. Beam broken > 3s = Storage Bin Full.
D11	**INPUT**	Metal Can Bin Optical Level	Active LOW (NPN NC)	Infrared through-beam across metal collection bin. Beam broken > 3s = Can Bin Full.
D13	**OUTPUT**	Internal Chamber LED Ring	5V Logic -> MOSFET	Illuminates scanning chamber during citizen intake; flashes green on accepted deposit.

CONTAINER SIZING LOGIC & REWARD POINT MATRIX

Classification	Optical Sensor States (D2, D7, D8)	Inductive (D5)	Height Window	Reward Credited
Small Plastic Bottle	D2 = HIGH, D7 = LOW, D8 = LOW	D5 = LOW (Non-metal)	Height ≤ 18 cm	**+5 Points** / 0.015 kg CO2
Medium Plastic Bottle	D2 = HIGH, D7 = HIGH, D8 = LOW	D5 = LOW (Non-metal)	18 cm < Height ≤ 26 cm	**+10 Points** / 0.025 kg CO2
Large Plastic Bottle	D2 = HIGH, D7 = HIGH, D8 = HIGH	D5 = LOW (Non-metal)	Height > 26 cm (max 32cm)	**+15 Points** / 0.040 kg CO2
Aluminium Can (UBC)	D2 = HIGH (Any height)	D5 = HIGH (Metal detected)	Standard UBC Dimensions	**+10 Points** / 0.050 kg CO2

ELECTRICAL ISOLATION MANDATE

The high-torque MG996R servo motor draw up to 2.5A peak during rapid gate actuation. Under no circumstances may servo VCC be powered from the Arduino 5V regulator. Servo VCC must connect directly to the external 5V 5A regulated buck rail with shared common ground.

3. Software Stack & Data Resiliency

WPF application lifecycle, local Microsoft SQL Server ledger schemas, and resilient Central Cloud synchronization.

LOCAL SQL SERVER DATABASE SCHEMAS

The kiosk maintains a persistent Microsoft SQL Server LocalDB database at C:\RVM\Data\RVM_LocalDB.mdf:

- dbo.Transactions: Records GUID, Timestamp, CitizenMobile, PlasticS/M/L, CanCount, TotalPoints, Co2SavedKg, and IsSynced (BIT).
- dbo.WalletAccounts: Tracks MobileNumber, LifetimePoints, AvailablePoints, and LastActiveDate for instant offline balance lookup.
- dbo.FeedbackRecords: Stores Rating (1-5), TagChips, Timestamp, and IsSynced.

CENTRAL CLOUD SYNCHRONIZATION PIPELINE

Managed by background worker CentralSyncService.cs:

- Executes FIFO upload batches of pending transactions (WHERE IsSynced = 0) via HTTPS REST API.
- Uses Bearer JWT Machine Token authentication for secure transmission.
- Upon HTTP 200 OK receipt from Central Cloud, local records are flagged IsSynced = 1.
- If network is offline, transactions buffer safely in local SQL Server without citizen disruption.

TELEMETRY & HEARTBEAT PROTOCOL (HEARTBEATSERVICE.CS)

Telemetry Field	Reporting Interval	Payload Description & Alert Triggering Thresholds
Machine Health Status	Every 60 Seconds	Reports ONLINE, MAINTENANCE, or FAULT based on serial and DB connectivity.
Storage Bin Capacity	Every 60 Seconds	Reports percentage full (0-100%). Triggers cloud SMS warning to caretaker when ≥ 85%.
Serial Hardware Link	Real-Time Event	Monitors USB COM port carrier detect. If disconnected, triggers critical dashboard alert.
Internal Chamber Temp	Every 300 Seconds	Monitors internal chassis temperature. Triggers warning if internal ambient > 48°C.
Sync Queue Depth	Every 60 Seconds	Counts un-synced transactions. Alert raised if queue exceeds 50 unsynced records.

OFFLINE FAULT TOLERANCE GUARANTEE

The RVM Desktop App is completely resilient to internet outages. All transactions, points credits, and citizen ratings are committed locally to ACID-compliant SQL Server storage before the UI proceeds. When 4G connectivity recovers, synchronization resumes automatically without data loss or duplicate point crediting.

STANDARD OPERATING PROCEDURES — PART 1 (OPERATIONS & MATERIAL RULES)

Mandatory Operating Procedures for Daily Startup, Pre-Flight Self-Test, and Acceptance Validation

SOP-RVM-OPS-01 Daily 07:00 AM

Daily Startup & Pre-Flight Self-Test

- Inspect external kiosk chassis, intake aperture, and touch screen for vandalism or damage.
- Verify AC 220V UPS connection; power on master rocker switch on the internal rear breaker panel.
- Confirm Windows 10 IoT executes launch-kiosk.ps1 and launches WPF full screen.
- Inspect the top diagnostics bar on primary screen:
  • SERIAL: CONNECTED 🟢
  • DB: OK 🟢
  • API: ONLINE 🟢
- Perform 1 test deposit using a clean 500ml PET bottle to verify gate servo sweep and point ledger update.
- Verify audio chime playback on successful acceptance.
- Sign daily morning pre-flight sign-off register.

SOP-RVM-OPS-02 Continuous Public

Material Rules & Anti-Cheat Validation

- **Accepted Materials:** Clear/colored PET plastic beverage bottles (0.25L to 2.5L), Aluminium UBC drink cans, and paper cups.
- **Strictly Prohibited:** Glass bottles (shatter hazard), motor oil or agrochemical containers, unemptied bottles containing liquid, crushed cans.
- **Single Feed Enforcement:** Containers must be inserted one-by-one, base first. Feeding multiple bottles simultaneously triggers anti-cheat rejection.
- **Liquid Contamination Check:** Optical refraction sensor detects residual liquid; gate rejects wet bottles to protect collection bin hygiene.
- **String/Foreign Object Detection:** If bottom sensor D2 remains continuously blocked for > 4.0 seconds, gate automatically enters lockout return mode.

DIAGNOSTICS BAR INDICATOR LOGIC

The top diagnostics bar dynamically updates every 1000ms. If SERIAL changes to 🔴 (RED), click the **RETRY** 🔄 button to re-poll COM ports. If API changes to 🟡 (YELLOW), system continues functioning in offline mode without disruption.

GLASS BOTTLE SAFETY PROHIBITION

Under NO circumstances should glass containers be allowed. The high-torque servo drop mechanism and drop chamber are engineered exclusively for non-shattering PET, HDPE, and aluminium UBC materials.

SOP-01 & 02 COMPLIANCE AUDIT CRITERIA

Station supervisors must verify that kiosks passing morning self-test have zero amber or red indicator pills. If any sensor test fails during initial test deposit, the machine must be placed in maintenance mode using the physical service key.

STANDARD OPERATING PROCEDURES — PART 2 (COLLECTION & PREVENTIVE CLEANING)

Protocols for Emptying Collection Storage Bins, Resetting Level Sensors, and Optical Maintenance

SOP-RVM-MAINT-03 At 85% Capacity or Bin Full

Storage Bin Servicing & Level Reset

- Type 888 on physical keypad to open Admin Security PIN prompt; pause active user sessions.
- Unlock bottom cabinet service door using the dual master security key.
- Pull out the 120L wheeled plastic collection trolley; tie off heavy-duty liner.
- Inspect plastic bin level sensor (D10) and metal bin sensor (D11) optical lenses. Wipe dust using dry microfiber cloth.
- Insert a fresh heavy-duty 120L transparent liner; ensure liner rim does not obscure the optical beam line.
- Slide trolley back into place and firmly latch bottom cabinet door.
- In Admin Panel -> Tab 4, click **"Reset Bin Full Counter"** to recalibrate capacity to 0%.
- Verify diagnostics bar clears the STORAGE BIN FULL 🔴 alert.

SOP-RVM-MAINT-04 Weekly (Every Friday)

Chamber Cleaning & Calibration

- Power down 12V sensor rail using internal service toggle switch.
- Wipe bottom (D2), mid (D7), and top (D8) IR lenses using 99% Isopropyl Alcohol (IPA) optical wipes.
- Clean inductive metal faceplate (D5) to remove adhesive residue and dust buildup.
- Clear dust particles from ultrasonic transducer emitter/receiver cones (D3/D4) using dry compressed air (< 30 PSI).
- Lubricate servo pivot hinge and nylon drop gate bushings with dry PTFE silicone spray.
- Restore 12V sensor power; press 33 on keypad to execute 20-sample baseline acoustic recalibration.
- Test 1 aluminium UBC can and 1 large PET bottle to verify classification accuracy.

OPTICAL SENSOR CLEANING & CARE REQUIREMENTS

Component	Approved Cleaning Agent	Technique & Prohibited Actions
Photoelectric Lenses (D2/7/8)	99% Isopropyl Alcohol (IPA)	Gently dab with lint-free optical swab. Do NOT use abrasive cloths, acetone, or window cleaners.
Ultrasonic Cones (D3/D4)	Clean Compressed Air (< 30 PSI)	Hold nozzle 15cm away; blow dust out. NEVER probe inside transducer meshes with sharp instruments.
Inductive Sensor (D5)	Mild Degreaser / Dry Cloth	Wipe flat sensor face clean of conductive beverage syrup. Ensure 2mm detection gap is unobstructed.
Touchscreen Glass	70% Ethanol / Screen Wipe	Spray onto microfiber cloth first; never spray liquid directly onto screen frame or bezel edges.

LINER BAG OBSTRUCTION HAZARD

When fitting replacement collection liners, caretakers must tuck bag excess firmly around trolley handles. If bag edges flap upwards into the D10/D11 infrared beam path, the kiosk will falsely report "Storage Bin Full" and lock out deposits.

STANDARD OPERATING PROCEDURES — PART 3 (SAFETY & DIAGNOSTIC TELEMETRY)

Emergency Jam Removal Protocols, Pinch Hazard Protection, and Hidden Telemetry Console Usage

SOP-RVM-SAFE-05 — Emergency Condition

Emergency Jam Clearance & Safety

- **PINCH HAZARD WARNING:** Immediately slap the Red Mushroom Emergency Stop button on the kiosk side panel.
- Verify all servo motors instantly de-energize and lock out.
- Unlock the upper chamber service hatch using the physical maintenance key.
- Using rubberized safety tongs or heavy work gloves, carefully remove stuck bottle, debris, or foreign objects.
- **NEVER insert bare hands or fingers into active servo gate mechanism.**
- Inspect acrylic chamber walls and optical sensor lenses for physical damage or scratches.
- Close and lock upper inspection hatch securely.
- Twist Red Emergency Stop button clockwise to release; system will automatically re-home servo gate to 0°.

SOP-RVM-DIAG-06 — Diagnostic Tool

Real-Time Telemetry Console

- On the kiosk hardware keypad, press **ESC** or key **8** to toggle the hidden real-time diagnostic terminal.
- The terminal appears as a semi-transparent black overlay displaying live serial COM packets:
  • IR_BEAMS: [D2:1, D7:0, D8:0]
  • ULTRA_DIST: 23.4 cm
  • INDUCTIVE: 0 (PET)
  • SERVO_POS: 0 DEG
- Verify memory allocation (WPF Managed Heap < 180 MB).
- Check SQL Server local connection pool status and API round-trip latency.
- Press **ESC** or key **8** again to dismiss the diagnostic console and restore citizen UI.

TELEMETRY CONSOLE SERIAL PACKET STRUCTURE

Packet Token	Expected Range / Values	Description & Fault Diagnosis
PING_OK	Periodic heartbeat	Confirms bidirectional serial link between Windows IoT host and ATmega328P.
SENS:[D2,D7,D8]	Binary (0 or 1 for each)	Real-time optical beam state. 1 = Beam clear; 0 = Beam interrupted by container.
IND:[0|1]	0 = Non-metal, 1 = Metal	Inductive sensor output. If permanently 1 with chamber empty, sensor requires recalibration.
DIST:[value]	04.0 cm to 80.0 cm	Ultrasonic distance. Baseline without bottle is typically 35.0 cm ± 2.0 cm.
GATE:[0|90|180]	Angle in degrees	Commanded servo position. 0° = Closed, 90° = Drop, 180° = Return/Reject.

EMERGENCY STOP ELECTRICAL DISCONNECT

The Emergency Stop circuit is hard-wired directly into the 12V/5V DC actuator bus. Pressing E-Stop mechanically breaks power to all motors independent of software state, guaranteeing fail-safe operator protection even during total software hangs.

STANDARD OPERATING PROCEDURES — PART 4 (DEMO MODE & IDLE EXPANSION)

Direct Demo Sequence Protocol (001), Hardware Key 0 Reservation, and Idle Geometry Standards

SOP-RVM-DEV-07 Testing / Offline Demo

Direct Demo Mode (001 Sequence)

- **Sequence 001:** When Arduino hardware is disconnected, typing the numeric sequence 0-0-1 on the keypad starts demo mode directly on the kiosk screen without dialogs.
- **Key 0 Reservation:** Key 0 is strictly reserved for physical hardware when connected. Pressing 0 triggers the physical gate opening and arms optical sensors.
- **Direct Simulation:** The kiosk transitions directly to PleaseInsert state with animated video overlay, simulated laser scanning, and point allocation.
- Allows field technicians to demonstrate complete citizen workflows to university administrators without physical bottles.
- Pressing ESC or completing the session cleanly returns to the home screen.

SOP-RVM-UI-08 Standby Feature (60s)

1-Minute Idle Expansion & Geometry

- **Idle Trigger:** Inactivity timer fires after exactly 60 seconds (1 minute) of zero touch or keypad input.
- **50% Video Expansion:** InstructionalVideoRowDef expands to 50% screen height (960px).
- **Contained Inner Zoom:** Inner video media element scales cleanly with ScaleTransform constrained inside container boundaries.
- **Clean Card Visibility:** "How to Use RVM" card and Ad Video section remain 100% visible with zero clipping.
- **Header Auto-Collapse:** Top header and hardware status bar collapse to 0 height.
- **Restoring Default Screen:** Pressing 0 or touching anywhere immediately zooms out and restores default screen.

GRID ROW DEFINITIONS & ZERO-OVERLAP ARCHITECTURAL STANDARDS

Grid Element	Default Screen Height	Idle Mode Height (60s)	Layout Behavior & Constraint
HeaderRowDef	Auto (~80px)	0 (Collapsed)	Fades opacity to 0 and collapses row height for full visual immersion.
InstructionalVideoRowDef	Auto (480px)	50* (960px = 50%)	Expands smoothly; contained inner zoom ensures video never spills outside.
HowToUseRowDef	Auto (~220px)	Auto (~220px)	Remains 100% visible below video with 16px bottom margin; zero overlap.
SignageVideoRowDef	* (Remaining height)	* (Remaining 740px)	Ad video continues looping cleanly without distortion or clipping.
BottomStatsRowDef	Auto (~140px)	0 (Collapsed)	Live counters collapse to make full room for the 50% video expansion.

HOTKEY QUICK-TESTING CONVENIENCE

Field engineers can instantly toggle Idle Mode standby expansion at any moment by pressing **Ctrl + I** on a service keyboard, without having to wait the full 60 seconds. Pressing **Ctrl + I** or pressing **0** returns immediately to the default screen.

STANDARD OPERATING PROCEDURES — PART 5 (OFFLINE SYNC & MEDIA ADS)

Data Ledger Reconciliation, Offline Queueing Protocols, and Digital Signage Management

SOP-RVM-NET-09 Network Failover

Offline Queueing & Data Reconciliation

- When 4G router disconnects or cloud API is unreachable, kiosk continues 100% normal operation.
- Transactions commit locally to SQL Server with column IsSynced = 0.
- Background worker retries every 30 seconds with exponential backoff.
- Upon network recovery, pending batches push automatically in chronological FIFO order.
- **Technician Manual Sync:** Open Admin Window (888) -> Tab 3 -> Click **"Sync Pending Records"**.
- Verify pending queue count drops to 0 and confirmation dialog reports sync success.

SOP-RVM-MKT-10 Weekly Media

Digital Signage Video Ads Management

- **Format Requirements:** MP4 container, H.264 video codec, AAC audio, 25/30 FPS.
- **Resolution:** 1080x720 (Primary Kiosk) / 1920x1080 (Secondary Landscape Display).
- Copy approved promotional media files into C:\RVM\Ads\ directory.
- Open Admin Window (888) -> Tab 6 (Advertisement & Video Signage).
- Select active playlist videos, set loop duration, and verify smooth preview playback.
- Click **"Save Playlist"**; both screens immediately apply the updated media reel.

OFFLINE RECONCILIATION VERIFICATION CHECKLIST

Verification Step	Expected Result	Corrective Action if Validation Fails
Local DB Record Count	SELECT COUNT(*) FROM Transactions	Matches total accepted sessions displayed on kiosk counter.
Unsynced Queue Depth	WHERE IsSynced = 0 returns 0 records	If > 0, verify internet connection and click "Sync Pending" in Admin Tab 3.
Cloud API Endpoint Ping	HTTP GET /api/health returns 200 OK	Check SIM card data balance, APN settings, and 4G antenna cable.
Wallet Balance Integrity	Citizen mobile balance matches cloud balance	Run reconciliation query: EXEC sp_ReconcileWallets.

BANDWIDTH OPTIMIZATION PROTOCOL

Sync payloads are compressed with GZIP and transmitted as lightweight JSON arrays. An entire day of 500 recycling deposits consumes less than 450 KB of cellular data, allowing dependable operation on modest 4G IoT SIM plans.

HARDWARE DIAGNOSTIC CODES & RAPID FAULT RESOLUTION MATRIX

Standard Field Corrective Action Procedures for Caretakers & Field Technicians

Fault Symptom / Indicator	Probable Root Cause	Immediate Corrective Action Procedure
SERIAL: DISCONNECTED 🔴 Hardware Connection Error	USB cable dislodged; Arduino unpowered; incorrect COM port assigned in config.txt.	1. Re-seat USB cable firmly into IPC port. 2. Check 5V power LED on Arduino board. 3. Click **RETRY** 🔄 on diagnostics bar to re-enumerate serial ports. 4. If still unresolved, verify COM port number in Windows Device Manager.
API: OFFLINE 🟡 Cloud Sync Deferred	Cellular 4G router offline; SIM data package expired; DNS server timeout.	1. Kiosk continues operating normally in offline mode. 2. Inspect 4G router status LEDs (Green = Connected, Red = SIM Fault). 3. Recharge SIM data pack if depleted. 4. Upon reconnection, background sync will auto-upload pending records.
STORAGE BIN FULL 🔴 Deposits Temporarily Locked	120L collection trolley at maximum capacity; or dust covering D10/D11 optical lenses.	1. Follow **SOP-03**: Unlock bottom door and replace liner bag. 2. Clean D10/D11 through-beam lenses with dry microfiber cloth. 3. Open Admin Window (888) -> Tab 4 -> Click "Reset Bin Full Counter".
BOTTLE STUCK / JAM ⚠️ Gate Jam Alarm	Oversized bottle (> 32cm); deformed can; foreign debris blocking drop gate.	1. Follow **SOP-05**: Press Red Emergency Stop button immediately. 2. Unlock upper service hatch. 3. Extract jammed item using safety tongs. 4. Release E-Stop; press 33 on keypad to execute acoustic calibration.
GATE SERVO UNRESPONSIVE ⚡ Servo Stalled	PWM signal lead loose on Pin D9; 5V 5A buck power rail voltage drop; stripped servo horn.	1. Measure voltage across 5V servo rail with digital multimeter (must be ≥ 4.85V). 2. Inspect D9 DuPont connector. 3. Replace servo motor if internal nylon/metal gears are stripped.
IDLE MODE NOT EXPANDING Standby Delay	Continuous touch input detected (ghost touch on dirty glass); timer paused.	1. Clean touchscreen surface with screen wipe to eliminate ghost touches. 2. Press **Ctrl + I** to manually verify expansion animation. 3. Verify idle timeout threshold in config.txt is set to 60 seconds.

HOTKEY MATRIX QUICK-SUMMARY

888: Admin Security PIN Prompt • 001: Direct Demo Testing • 0: Hardware Trigger / Idle Exit • 33: Chamber Acoustic Calibration • 8 / ESC: Telemetry Terminal Console • Ctrl + I: Idle Expansion Toggle.

10. Maintenance Schedule & Field Sign-Off

Mandatory service frequencies, preventive maintenance tasks, and official technician audit sign-off ledger.

Cadence	Mandatory Preventive Action	Inspection Standard & Criteria
Daily (07:00 AM)	Pre-flight diagnostics, test deposit, bin fullness check.	Zero red indicators; clean touchscreen; bin capacity < 85%.
Weekly (Friday)	Optical lens cleaning (IPA), dust blowing, acoustic calibration (33).	Clean lenses (D2/7/8/10/11); distance reading 35cm ± 2cm.
Monthly (1st)	Servo gear inspection, PTFE lubrication, SQL Server database maintenance.	Smooth 90° gate sweep; DB log truncation; queue depth = 0.
Quarterly	AC UPS battery capacity test, chassis exhaust fan cleaning, screw torque check.	UPS runtime > 35 mins on battery; all wiring terminals torqued.

FIELD SERVICE & ENGINEERING SIGN-OFF AUDIT LOG

Date	Time	Technician Name	Machine ID	Action / Maintenance Performed	Status	Signature
__/__/2026	__:__ AM	____________	RVM-001	Morning startup, test deposit, diagnostics check	**PASSED**	____________
__/__/2026	__:__ AM	____________	RVM-001	Storage bin emptied, fresh 120L liner installed	**RESET OK**	____________
__/__/2026	__:__ AM	____________	RVM-001	Weekly optical IPA wipe, acoustic calibration	**CALIBRATED**	____________
__/__/2026	__:__ AM	____________	RVM-001	Monthly servo maintenance, queue audit	**VERIFIED**	____________
__/__/2026	__:__ AM	____________	RVM-001	____________________________	________	____________

LEAD FIELD ENGINEER CERTIFICATION

I hereby certify that this Reverse Vending Machine has been serviced, tested, and calibrated in accordance with the specifications in SOP-RVM-ENG-2026-V3.2.

EMERGENCY HOTLINE & TECHNICAL DISPATCH

Field Engineering Central Dispatch: **0800-RECYCLE (0800-73292)**
Engineering Support Portal: **isprvm.binishaqsoft.com/eng**